

Ayush Girhay

B.Tech. III Year, Civil Engineering
Indian Institute of Technology Roorkee

☎ +91-9860334977 ✉ ayushgirhay@gmail.com

Profile

AI/ML, Large Language Models, Computer Vision, NLP, Data Science, Deep Learning, Web-Development, Data Structures and Algorithms

AIR-6707 JEE Advanced 2023 **AIR-973** JEE Mains 2023

Education

Indian Institute of Technology, Roorkee 2023 – Present

B.Tech., Civil Engineering

Insight JR Collage (MSBSHSE) 2021 – 2022

Intermediate (Class XII)

Experience

Participant — GenAI-Powered Data Analytics Job Simulation Jun 2025 – Jul 2025

Forage — Tata Group

Completed a virtual simulation covering exploratory data analysis, risk profiling, predicting delinquency using AI models, business reporting/storytelling, and designing AI-driven collection strategies.

Participant — Software Engineering Job Simulation Jun 2025 – Jul 2025

Forage — Accenture

Completed virtual experience involving software architecture, secure coding practices, programming exercises, agile development, and testing modules.

Projects

Creating a Large Language Model Jun 2025 – Present

Built a basic LLM from scratch using Python by implementing tokenization, embeddings, attention, transformer blocks, self-attention, positional encoding, and training pipelines.

AI-Powered Event and Media Management Platform May 2026 – June 2026

Built a scalable full-stack event management system featuring event scheduling, ticketing, attendee tracking, and analytics dashboards. Designed responsive user interfaces and optimized backend services for seamless event operations.

Dynamic Tariff Optimization for EV Charging Networks Sep 2023 – Nov 2023

Built an AI-driven dynamic pricing engine for EV charging networks, leveraging predictive analytics and autonomous decision-making to optimize tariffs based on demand, occupancy, and usage patterns.

Dynamic Pricing for Urban Parking Lots May 2025 – Jul 2025

Consulting & Analytics Club IIT Guwahati x Pathway

Built a real-time parking pricing system using demand, traffic, and competition signals with Python, NumPy, Pandas, Pathway, and Bokeh on data from 14 parking lots.

House Price Prediction Dec 2025 – Jan 2026

Built a multimodal ML model using tabular data and satellite imagery for house price prediction using PCA, XGBoost, SHAP, and Grad-CAM.

Technical Skills

Languages: Python, C++, SQL, HTML, CSS, JavaScript, Bootstrap, React.js, ROS2

Data Analytics: EDA, Pandas, NumPy, Matplotlib, Data Visualization

Software Packages: ArcGIS, QGIS, Solidworks, Fusion360

Tools: Jupyter Notebook, Google Colab, scikit-learn, MySQL, Tableau, Excel, Git, GitHub, VS Code, Postman

Positions of Responsibility

Captain — Hockey IIT Roorkee July 2023 – Present

- Proficiency holder in National Sports Organisation (NSO) and member of the institute's hockey team. Contributed to the team's success through dedication, discipline and passion for the game

Team member — Models and Robotics Section (MARS), IIT Roorkee Jun 2025 – Present

- Core Member of MARS (IIT Roorkee), contributing to innovative robotics projects involving interdisciplinary collaboration. Worked on CAD design using SolidWorks and Fusion 360, along with AI/ML integration and robot prototyping. Coordinated closely with electronics and programming teams to ensure smooth development, testing, and implementation of robotic systems.